

Ian Leung

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EDUCATION

University of Waterloo

Bachelor of Computer Science & Statistics

Sep 2025 – May 2029

GPA: 3.8/4.0

- President's Scholarship of Distinction (\$5000), Undergraduate Achievement Award (\$1500)
- Relevant Coursework: Advanced: Object-Oriented Programming, Sequential Programming, Functional Programming, Interpreters; also Computer Architecture, Calculus, Linear Algebra, Data Analytics

EXPERIENCE

Roblox

Jun 2026 – Aug 2026

Software Engineer Intern (Network Reliability) | Go, Python

San Mateo, CA

- Worked alongside network engineers to automate network deployments, pushing to **5000+ production devices** in global datacenters with a GitHub Actions CI/CD pipeline equipped with staged expansion and health checks
- Orchestrated Temporal Workflows with LLM analysis & Slack alerting for safe, autonomous rollout of configuration changes, saving **400+ engineering hours** annually and identifying issues on 20+ network devices
- Identified and fixed a resource leak in an internal network service, recovering 7% CPU utilization across devices

WQC

Jan 2026 – Present

Software Engineer | Rust, Python, Docker

Waterloo, ON

- Engineered portal.waterlooquantclub.com, a full-stack club recruitment & event portal hosted on AWS for **800+ users**
- Launched a Grafana observability platform with 100+ metrics, including membership and infrastructure monitoring
- Configured Alertmanager to send automatic Discord alerts for critical service failures, maintaining a **>99.6% uptime**
- Built and deployed 2v2bluff, a real-time multiplayer Rust/WebSocket card game supporting four concurrent players

Sunnybrook Research Institute

Jul 2025 – Aug 2025

Software Research Intern | Python

Toronto, ON

- Built an event tracing, playback, and logging tool that records and simulates clinician workflows in Focused Ultrasound software, surfacing 5+ defects and edge cases in real usage patterns
- Automated regression testing by generating synthetic JSON event flows and validating results against recorded baselines, **cutting testing time by 91%**

PROJECTS

Jevis: Hack The North 2026 Winner (\$2000) | Python, DOM, JEV, OpenAI

- Engineered an agentic harness for browser use using the JEV System One Model, **decreasing token cost by 250x** compared to GPT-6 Astra
- Sped up queries on Walmart from **5 minutes to 44 seconds** using a DOM-based approach and orchestrator models
- Built a caching layer for repeated agent tasks via precomputed instruction paths, achieving a 15% cache hit rate

Pharmagotchi: AI Pharmacy Companion App | Kotlin, Android Studio, Figma, OpenRouter

- Built a Tamagotchi-inspired Android companion app to track health & medications and connect users with physicians
- Integrated an OpenRouter-powered chat assistant with Retrieval-Augmented Generation (RAG) that parses health conditions and medications into structured JSON vitals and provides context-aware medication guidance
- Connected Fitbit tracking devices through the Google Health API, centralizing health data for seamless monitoring

PeerAssist: Peer Review Platform | Java, Supabase, Apache PDFBox, Maven

- Created a full-stack Java desktop application for 20+ students to upload, review, and grade peers' academic work
- Integrated a Supabase backend for authentication and data storage, enforcing Row-Level Security policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with Apache PDFBox and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via Maven shade plugin

SKILLS

Languages Python, Java, C++, C#, Go, Kotlin, TypeScript, SQL, Bash

Technologies Temporal, gRPC, REST, PyTorch, Pandas, NumPy, OpenCV, PostgreSQL, React, Node.js

Tools Git, Docker, AWS, GitHub Actions, Grafana, OpenRouter, Supabase, Android Studio